Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 1

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KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · convert *(panoni)*

A Fractions and decimals ●○○

Remember: A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. To go back, use place value: $0.6 = \frac{6}{10} = \frac{3}{5}$.

1. Write each fraction as a decimal.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{2}{5}$ = _____ **d)** $\frac{7}{10}$ = _____

e) $\frac{1}{8}$ = _____ **f)** $\frac{9}{20}$ = _____ **g)** $\frac{3}{25}$ = _____ **h)** $\frac{5}{8}$ = _____

2. Write each decimal as a fraction in its simplest form.

a) 0.3 = _____ **b)** 0.5 = _____ **c)** 0.25 = _____ **d)** 0.8 = _____

e) 0.45 = _____ **f)** 0.06 = _____ **g)** 0.125 = _____ **h)** 0.72 = _____

B Decimals and percentages ●○○

Remember: Per cent means "out of 100". Multiply a decimal by 100 to get a percentage, and divide a percentage by 100 to get a decimal.

3. Write each decimal as a percentage.

a) 0.45 = _____ **b)** 0.7 = _____ **c)** 0.08 = _____ **d)** 0.125 = _____

e) 1.5 = _____ **f)** 0.006 = _____ **g)** 0.99 = _____ **h)** 2.4 = _____

4. Write each percentage as a decimal.

a) 25% = _____ **b)** 60% = _____ **c)** 7% = _____ **d)** 12.5% = _____

e) 130% = _____ **f)** 0.5% = _____ **g)** 99% = _____ **h)** 240% = _____

C Fractions and percentages ●●○

5. Write each fraction as a percentage.

a) $\frac{1}{2}$ = _____ **b)** $\frac{3}{4}$ = _____ **c)** $\frac{2}{5}$ = _____ **d)** $\frac{3}{10}$ = _____

e) $\frac{7}{20}$ = _____ **f)** $\frac{9}{25}$ = _____ **g)** $\frac{3}{8}$ = _____ **h)** $\frac{7}{50}$ = _____

6. Write each percentage as a fraction in its simplest form.

a) 50% = _____ **b)** 25% = _____ **c)** 40% = _____ **d)** 90% = _____

e) 15% = _____ **f)** 64% = _____ **g)** 5% = _____ **h)** 12.5% = _____

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D Putting them in order ●●○

> **Remember:** Change every number to the same form (decimals are usually easiest), then compare.

7. Write each list in order, smallest first.

a) $0.4, \quad 35\%, \quad \frac{3}{8}$

b) $\frac{3}{5}, \quad 0.65, \quad 62\%$

c) $7\%, \quad 0.7, \quad \frac{7}{20}$

d) $\frac{1}{3}, \quad 0.3, \quad 35\%$

E Percentage of an amount ●●○

> **Remember:** Change the percentage to a decimal and multiply: 35% of 80 is 0.35×80.

8. Work out.

a) 50% of 84

b) 10% of 230

c) 25% of 48

d) 20% of 65

e) 75% of 120

f) 5% of 360

g) 30% of \$250

h) 15% of \$80

i) 12% of 400 g

j) 35% of 60 kg

k) 8% of 150

l) 120% of 45

F One amount as a percentage of another ●●●

> **Remember:** Write the two amounts as a fraction, then multiply by 100.

9. Answer each question.

a) What percentage of 50 is 12?

b) What percentage of 40 is 18?

c) What percentage of 250 is 45?

d) What percentage of 20 is 7?

e) Kiri scored 27 out of 60 in a test. What percentage is that?

f) A shop sold 36 of its 150 pies before noon. What percentage is that?

★ Extension

10. Mere scored 32 out of 40 in a maths test and 27 out of 36 in a science test. In which test did she do better? Use percentages.

11. 30% of a number is 45. Find the number.

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Fractions, Decimals and Percentages

Name _____ Date _____ No calculator Version 2 Scan to practise this topic online

KEY WORDS fraction *(hautanga)* · decimal *(tau ā-ira)* · percentage *(ōrau)* · convert *(panoni)*

A Fractions and decimals ●○○

Remember: A fraction is a division: $\frac{3}{4} = 3 \div 4 = 0.75$. To go back, use place value: $0.6 = \frac{6}{10} = \frac{3}{5}$.

1. Write each fraction as a decimal.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{3}{5}$ = _____ **d)** $\frac{9}{10}$ = _____

e) $\frac{3}{8}$ = _____ **f)** $\frac{7}{20}$ = _____ **g)** $\frac{6}{25}$ = _____ **h)** $\frac{7}{8}$ = _____

2. Write each decimal as a fraction in its simplest form.

a) 0.7 = _____ **b)** 0.2 = _____ **c)** 0.75 = _____ **d)** 0.4 = _____

e) 0.35 = _____ **f)** 0.08 = _____ **g)** 0.375 = _____ **h)** 0.64 = _____

B Decimals and percentages ●○○

Remember: Per cent means "out of 100". Multiply a decimal by 100 to get a percentage, and divide a percentage by 100 to get a decimal.

3. Write each decimal as a percentage.

a) 0.35 = _____ **b)** 0.9 = _____ **c)** 0.03 = _____ **d)** 0.375 = _____

e) 1.2 = _____ **f)** 0.004 = _____ **g)** 0.81 = _____ **h)** 3.5 = _____

4. Write each percentage as a decimal.

a) 45% = _____ **b)** 30% = _____ **c)** 9% = _____ **d)** 37.5% = _____

e) 160% = _____ **f)** 0.2% = _____ **g)** 88% = _____ **h)** 310% = _____

C Fractions and percentages ●●○

5. Write each fraction as a percentage.

a) $\frac{1}{4}$ = _____ **b)** $\frac{1}{5}$ = _____ **c)** $\frac{4}{5}$ = _____ **d)** $\frac{7}{10}$ = _____

e) $\frac{11}{20}$ = _____ **f)** $\frac{8}{25}$ = _____ **g)** $\frac{5}{8}$ = _____ **h)** $\frac{9}{50}$ = _____

6. Write each percentage as a fraction in its simplest form.

a) 75% = _____ **b)** 20% = _____ **c)** 60% = _____ **d)** 30% = _____

e) 35% = _____ **f)** 48% = _____ **g)** 4% = _____ **h)** 37.5% = _____

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D Putting them in order ●●○

> **Remember:** Change every number to the same form (decimals are usually easiest), then compare.

7. Write each list in order, smallest first.

a) $0.6, \quad 65\%, \quad \frac{5}{8}$

b) $\frac{2}{5}, \quad 0.45, \quad 42\%$

c) $9\%, \quad 0.9, \quad \frac{9}{20}$

d) $\frac{2}{3}, \quad 0.6, \quad 65\%$

E Percentage of an amount ●●○

> **Remember:** Change the percentage to a decimal and multiply: 35% of 80 is 0.35×80.

8. Work out.

a) 50% of 96

b) 10% of 470

c) 25% of 64

d) 20% of 85

e) 75% of 160

f) 5% of 240

g) 30% of \$350

h) 15% of \$60

i) 12% of 300 g

j) 45% of 80 kg

k) 6% of 250

l) 130% of 40

F One amount as a percentage of another ●●●

> **Remember:** Write the two amounts as a fraction, then multiply by 100.

9. Answer each question.

a) What percentage of 50 is 16?

b) What percentage of 40 is 14?

c) What percentage of 250 is 35?

d) What percentage of 20 is 9?

e) Kiri scored 33 out of 60 in a test. What percentage is that?

f) A shop sold 48 of its 150 pies before noon. What percentage is that?

★ Extension

10. Mere scored 18 out of 24 in a maths test and 28 out of 35 in a science test. In which test did she do better? Use percentages.

11. 30% of a number is 36. Find the number.

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